Enhancing the Spatial Resolution of Hyperspectral Images Combining High-Accuracy Surface Modeling and Subpixel Unmixing

Jia Chen, *Member, IEEE*, Jun Li, *Fellow, IEEE*, and Paolo Gamba, *Fellow, IEEE*

***Abstract*— Hyperspectral sensors can rapidly acquire high-quality spectral data, very useful for urban monitoring applications. Unfortunately, their spatial detail is not fine enough, and methods to enhance this resolution are required. However, conventional super-resolution (SR) methods for multispectral data do not match the requirements needed to maintain high spectral fidelity. Therefore, this article proposes a hyperspectral SR method that combines subpixel mapping and interpolation, and whose main aim is to enhance urban monitoring. This method aims to guarantee spectral quality and minimize computational time through a high-precision surface interpolation method based on curve theory. Moreover, unmixing-based subpixel mapping is exploited to introduce unmixing information. Finally, using wavelet transforms, the method integrates the effective information from the two previous approaches, obtaining urban hyperspectral images with enhanced spatial details and spectral fidelity. This method has been subjected to a comprehensive experimentation, affirming that the proposed method surpasses the current state-of-the-art SR results in terms of performance and effectiveness.**

***Index Terms*— Autoencoder, graph network, hyperspectral imaging, multitask learning, remote sensing, superpixel segmentation, unmixing.**

I. Introduction

URBAN hyperspectral data [1], due to their rich spectral information, play a very important role in urban planning, environmental monitoring, and disaster control. These data can provide detailed information about urban structure and environmental changes, which are crucial for understanding and managing urban ecosystems. However, due to technological limitations, the spatial resolution of these images is low. This greatly limits the practicality of hyperspectral data. Image super-resolution (SR) can greatly improve the spatial details of images through technical means, allowing for more precise and detailed urban analysis. SR plays an important in other fields such as the identification of objects in the locations [2], detecting small objects [3], [4], medical diagnostics [5], improved analysis in remote sensing images [6], astronomy [7], and forensics [8].

SR techniques, as delineated by Glasner et al. [9], can be broadly categorized into two types: learning-based and interpolation-based methods. Learning-based SR methods, particularly those harnessing the power of deep learning, have garnered significant attention for their superior performance in recent years. These methods involve training deep neural networks on a dataset comprising pairs of low-resolution and high-resolution images to effectively learn the transformation from low to high resolution.

Among the learning-based approaches, convolutional neural networks (CNNs) have been predominant. Techniques such as SRCNN [10], very deep SR (VDSR) [11], enhanced deep SR network (EDSR) [12], and efficient subpixel CNN (ESPCN) [13] use various CNN architectures to upscale images progressively or directly predict high-resolution outputs from low-resolution inputs. The success of these models lies in their ability to capture and use complex image features efficiently.

Another groundbreaking approach involves the use of generative adversarial networks (GANs). Methods such as SRGAN [14], ESRGAN [15], and real-ESRGAN [16] leverage the adversarial training framework, wherein a generator creates high-resolution images and a discriminator evaluates them. This competition drives the generator to produce increasingly realistic and high-quality super-resolved images, pushing the boundaries of what can be achieved with SR.

On the other hand, interpolation-based SR techniques, identified as one of the earliest SR methods [17], operate on a simpler principle. They enhance image resolution by estimating new pixel values in low-resolution images using mathematical interpolation methods. Techniques such as nearest-neighbor, bilinear, bicubic interpolation, and thin plate spline (TPS) [18] are used to fill in the gaps in low-resolution images, aiming to create smoother and more visually appealing high-resolution versions. Despite being less sophisticated than learning-based methods, interpolation techniques are still used due to their simplicity and effectiveness in certain applications.

Learning-based methods have been successful in producing high-quality SR images, capturing fine details and textures more effectively than traditional methods. The strength of learning-based approaches lies in their ability to leverage complex models to learn from vast amounts of data, thereby

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Jia Chen was with the Department of Electrical, Biomedical and Computer Engineering, University of Pavia, 27100 Pavia, Italy. He is now with the School of Earth Resources, University of Geosciences, Wuhan 430079, China.

Jun Li is with the School of Earth Resources, University of Geosciences, Wuhan 430079, China.

Paolo Gamba is with the Department of Electrical, Biomedical and Computer Engineering, University of Pavia, 27100 Pavia, Italy (e-mail: paolo.gamba@unipv.it).

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generating results that often surpass what is possible with simpler algorithms. Despite their effectiveness in dealing with diverse scenarios and capturing intricate details necessary for urban imagery, these methods have several drawbacks.

First, learning-based SR methods require substantial computational resources. The training of deep neural networks is computationally intensive and often necessitates high-performance GPU hardware, which can be prohibitively expensive and energy-consuming. This makes the approach less feasible for real-time applications or for use in environments with limited hardware capabilities. Second, these methods are prone to overfitting, especially when the training data are not sufficiently diverse or large enough to represent the variability encountered in real-world scenarios. Overfitting results in models that perform well on training data but fail to generalize to new, unseen data, which can be particularly problematic in dynamic urban environments where the spectral characteristics can vary widely.

Interpolation-based methods are instead computationally efficient and straightforward to implement. They do not require extensive training and sophisticated hardware that learning-based methods do, making them more accessible for quick applications and on systems with limited computational power. However, these methods often lack the ability to recreate high-frequency details accurately. Since interpolation techniques essentially estimate pixel values based on the spatial relationships of surrounding pixels, they fail in reconstructing complex textures and the fine details of urban hyperspectral images. This limitation often leads to artifacts or blurred images in the SR output. Artifacts such as ringing or aliasing degrade the quality of the super-resolved image, while blurring mask important features necessary for tasks such as feature recognition and classification in urban planning and monitoring.

In summary, while learning-based methods offer superior detail and textural accuracy, their practical deployment is hampered by high computational demands and potential overfitting issues. Interpolation-based methods, while beneficial for their simplicity and efficiency, fall short in delivering the high-fidelity results required for detailed urban hyperspectral analysis. Both the methods need further refinement and adaptation to overcome these challenges in urban hyperspectral imaging applications.

To overcome the aforementioned problems, this article proposes a hyperspectral SR method that combines subpixel mapping and interpolation. This method aims to ensure spectral information and minimize time consumption through a high-accuracy surface modeling (HASM) based on curve theory. In addition, using subpixel mapping to introduce unmixing information, it seeks to compensate for spatial details. Finally, using wavelet transformation, the method integrates the effective information from both the approaches, obtaining urban hyperspectral images with enhanced spatial detail and spectral fidelity.

The primary contributions of this method are therefore as follows.

1) *Integration of HASM in Hyperspectral Data Processing:* By adopting HASM, which is based on curve theory, the method enhances the precision of spatial detail enhancement while preserving the integrity of spectral information. This ensures that the spectral characteristics of the urban landscape are accurately maintained, which is crucial for applications such as environmental monitoring and material identification.
2) *Incorporation of Subpixel Mapping for Enhanced Unmixing:* The use of subpixel mapping introduces an additional layer of detail by incorporating unmixing information in the overall SR approach. This step is vital for accurately capturing complex spatial structures of urban environments, where different materials and surfaces may exist within a single pixel, thereby providing a more nuanced reconstruction of the scene.
3) *Utilization of Wavelet Transformation:* Using wavelet transformation allows the method to effectively merge the information captured by both the HASM and subpixel mapping techniques. This fusion step not only enhances the resolution but also ensures that the transitions between different resolutions are smooth and artifact-free. The result is an hyperspectral image that boasts improved spatial detail and spectral fidelity, making it more useful for detailed analytical tasks.

These contributions demonstrate the effectiveness and applicability of our proposed method in hyperspectral image analysis.

The rest of this article is organized as follows. The introduction of research related to autoencoding SU is delineated in Section II. In Section III, the proposed method is presented in detail, and the main flowchart is given. The experiments and results are shown in Section IV. The sensitivity analysis of some parameters is performed in Section V. The conclusion is drawn in Section V.

II. Related Work

SR techniques have become a focal point in image processing, aiming to reconstruct high-resolution images from their low-resolution counterparts. These techniques are categorized mainly into learning-based and interpolation-based approaches, each contributing significantly to the field's evolution.

A. Learning-Based SR

The development of learning-based SR has been predominantly influenced by the advent of deep learning, with CNNs playing a crucial role. The SRCNN by Dong et al. [10] marked a significant breakthrough by demonstrating the effectiveness of CNNs in learning end-to-end mappings from low- to high-resolution images. This was followed by the introduction of the VDSR model by Kim et al. [11], which used deeper networks to learn residuals between low- and high-resolution images, significantly enhancing the reconstruction quality. DnCNN [19] learns to directly forecast high-frequency residual between LR and HR images. IRCNN [20] suggests a series of denoisers related to CNN which could be used mutually for various lower level computer vision tasks. EBSR [21] separates the multiframe SR problem into three parts: alignment, fusion, and reconstruction. WMCNN [22]

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is based on wavelet analysis by training CNNs for multiscale representations. Residual learning network (RLN) [23] architectures, as opposed to linear networks, incorporate skip connections to prevent gradient vanishing and enable the design of deeper networks. The recent RLNs [24], [25], [26] are categorized on the basis of the number of phases used in the networks. An individual network consists of a single-stage design. EDRN [12], for SR, modifies a ResNet structure originally proposed for image classification. DRLN [27] has an l1 loss function similar to RCAN [28] for training network. The settings for training are the same as RCAN with the quantity of epochs, optimizer, and training patches.

Further advancements were made with the enhanced EDSR [12] and the ESPCN [13], which refined performance by optimizing network structures and upscaling methods. EDSR, in particular, removed unnecessary modules in conventional CNNs, significantly enhancing performance without additional computational complexity. ESPCN introduced an efficient subpixel convolution layer that directly upscales the resolution of the input image at the network's end, thereby reducing computational costs and improving image quality.

GANs have also significantly impacted SR through adversarial training. SRGAN by Ledig et al. [14] was pioneering, introducing a perceptual loss function that focuses on achieving high perceptual quality. Subsequent works such as ESRGAN [15] and real-ESRGAN [16] improved the realism and detail in the images by modifying network architectures and training strategies, better capturing texture details, and reducing artifacts. SRFeat [29] is another GAN-based SR algorithm focusing on natural perception using feature discrimination. It uses an additional discriminator to differentiate between synthetic and actual images.

B. Interpolation-Based SR

Interpolation-based SR techniques are some of the oldest methods in image resolution enhancement. They work by estimating new pixel values based on the known values of surrounding pixels. Nearest-neighbor interpolation, bilinear interpolation, and bicubic interpolation [30] are traditional forms, with bicubic generally providing the best balance between processing time and output quality.

Advanced interpolation methods such as TPS [18] offer more sophisticated options. TPS models the intensity surface of the image using splines under tension, providing smoother and more flexible transformations. This method is particularly effective in handling geometric distortions in addition to improving resolution. In this approach, the use of permanent function kernels, local variance coefficients [30], and adaptive structure kernels [31], [32] is common. These methods are straightforward and dependable for SR, yet they can lead to visual artifacts under certain conditions, especially with increased scaling factors. A detailed examination of these SR techniques and their constraints is discussed in [33]. Furthermore, Zhang et al. [34] developed a method that transforms a low-resolution image into a high-resolution image, ensuring the preservation of textural and structural details. Initially, they create a novel bivariate rational fractal interpolation model and analyze its analytical properties.

C. Comparative Analysis

Comparatively, while learning-based methods are capable of generating high-quality images, they require significant computational resources and extensive training datasets. Interpolation-based methods offer quicker and simpler solutions but typically at the cost of lower enhancement in resolution. The choice between these methods largely depends on specific application requirements, including available computational resources, desired image quality, and the nature of the input data.

In conclusion, SR remains a vibrant area of research with continuous developments that improve the performance and applicability of these techniques in various real-world scenarios.

III. Methodology

To achieve rapid acquisition, spectral fidelity, and rich spatial detail in urban hyperspectral imaging, this article introduces an SR technique that synergistically combines the advantages of two distinct approaches. The proposed method consists of three primary steps, each designed to enhance the final image quality while ensuring computational efficiency.

Initially, a baseline image is acquired through HASM for spatial interpolation. This step uses advanced curve theory to model the surface details with high precision, effectively setting a robust foundation for enhancing spatial resolution. HASM is particularly advantageous for its ability to preserve essential spectral characteristics while preparing the image for further refinement.

Subsequently, enriched spatial information is integrated using subpixel mapping. This technique refines the spatial resolution by dispersing pixel values into subpixel components based on their spectral signatures. It allows for a more granular representation of urban textures and features, which are crucial for applications such as urban planning and environmental monitoring. By applying subpixel mapping, the method enhances the detail and clarity of structural elements within the urban landscape.

Finally, wavelet transformation is used to amalgamate the effective information from both the methodologies to generate the target image. This transformation facilitates the fusion of the high-frequency details from the subpixel mapping with the low-frequency components preserved by HASM. As a result, the final image benefits from an optimized balance of spatial and spectral accuracy. The wavelet transformation also aids in reducing noise and artifacts, thereby producing a cleaner and more precise hyperspectral image.

The algorithmic workflow is comprehensively illustrated in Fig. 1. This figure provides a visual guide to the sequential application of these methods, highlighting how each contributes to the overall effectiveness of the proposed SR technique. By integrating these approaches, the method not only achieves enhanced resolution but also ensures that the enhanced images are of high quality, suitable for detailed analysis and decision-making in urban management.

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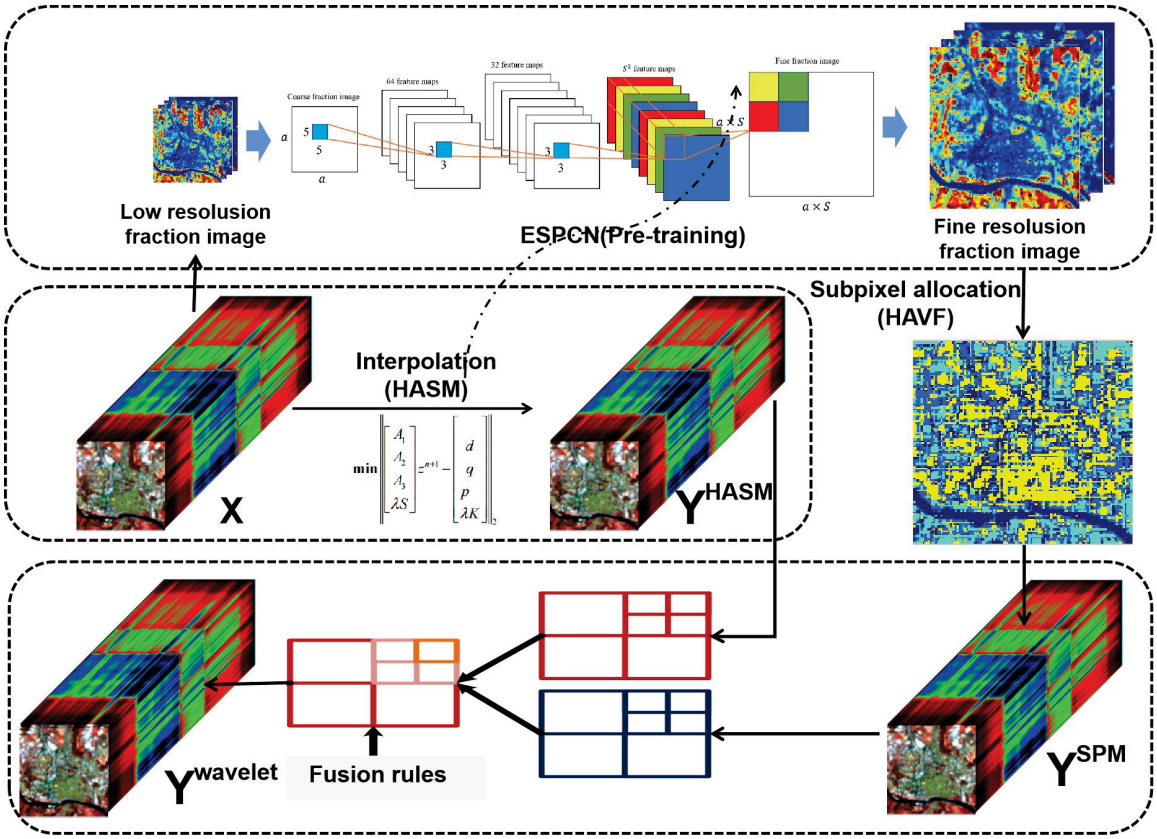


Fig. 1. Flowchart of the algorithm model.

A. High-Accuracy Surface Modeling

HASM [35] is an advanced interpolation technique designed to generate high-precision surface models from scattered data points. Originally developed for applications in geosciences and environmental studies, HASM has been adapted for enhancing spatial resolution in urban hyperspectral imaging while preserving spectral integrity. This method is rooted in the principles of minimal energy surfaces and uses partial differential equations (PDEs) to achieve optimal surface modeling.

The core of HASM lies in its ability to model surfaces by minimizing the energy associated with a thin plate under tension, akin to a physical elastic sheet stretched to cover the data points. The mathematical formulation is based on the theory of minimal energy surfaces, which is expressed through a PDE as follows:

$$\Delta^2 z - \alpha_1 \Delta z = 0 \tag{1}$$

where Δ^2 is the biharmonic operator, Δ is the Laplacian operator, z represents the surface elevation, and α_1 adjusts the weight of the Laplacian operator Δz in the equation, which affects the balance between surface smoothness and adherence to the data. This equation ensures that the resulting surface is not only smooth but also accurately conforms to the underlying data points.

In practical terms, HASM calculates the surface $z(x, y)$ that minimizes the functional

$$E(z) = \int \left[(\Delta z)^2 + \alpha_2 (\nabla z)^2 \right] dA \tag{2}$$

where $E(z)$ represents the total energy of the surface, and ∇z signifies the gradient of the surface, indicating the rate of change in height across the surface. α_2 determines the weight of the surface gradient term $(\nabla z)^2$, controlling the balance between smoothness and data fidelity. The integral is computed over the area A which covers all the data points. The minimization of this energy functional is typically achieved using numerical methods such as finite difference or finite element methods, which discretize the surface into a grid or mesh and solve the resulting linear or nonlinear system of equations.

In the context of urban hyperspectral imaging, HASM can be particularly effective for reconstructing detailed urban landscapes from sparse or incomplete data. By ensuring that the interpolated surfaces maintain both spatial continuity and a high level of detail, HASM supports the accurate characterization of urban features, which is critical for applications such as urban planning, environmental monitoring, and infrastructure management.

The adaptation of HASM to hyperspectral imaging allows for the maintenance of spectral fidelity in the process of spatial enhancement, thereby ensuring that the enhanced images are not only more detailed but also true to the original spectral characteristics of the urban landscape.

Assuming that a hyperspectral image is transformed into a surface model based on the fundamental laws of surface theory, a spatial surface is completely determined by the first-order and second-order fundamental quantities. Under controlled conditions, a unique solution exists, and the

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corresponding mathematical expression simplifies to

$$\min \left\| \begin{bmatrix} A \\ B \\ C \\ \gamma S \end{bmatrix} Z^{n+1} - \begin{bmatrix} d \\ p \\ q \\ \gamma k \end{bmatrix} \right\|_2 . \tag{3}$$

It is important to note that the interpolation points S are obtained through TPS [18]. A, B, and C represent the constraints of the surface. d, q, p, and k are the parameters of these constraints. Z^{n+1} represents the surface solution obtained after $(n+1)$ iterations. It is the discretized surface representation of z in (1) and (2).

In addition, endmembers and their abundances can be derived from the original low-resolution hyperspectral image X using unmixing methods. In this case, the multitask autoencoder hyperspectral unmixing model (MAHUM) [36] is used for unmixing. The derived endmember abundance maps are then subjected to a subpixel mapping process where special consideration is given to the interpolation magnification mode. This ensures that the spatial details are enhanced without altering the original material properties encoded by the endmembers.

In designing the subpixel mapping method, a technique analogous to SR, specifically the ESPCN [13], is used. ESPCN is renowned for its ability to upsample low-resolution images to higher resolutions effectively. The network achieves this by learning an end-to-end mapping between low- and high-resolution images. A significant feature of ESPCN is its use of subpixel convolution layers, which help in reconstructing the high-resolution output from the low-resolution input more efficiently than traditional interpolation methods. This approach not only preserves the high-frequency details but also reduces the computational complexity and improves the speed of the SR process.

B. Efficient Subpixel Convolutional Neural Network

The ESPCN represents a significant advancement in the field of image SR. Unlike traditional methods that rely on interpolation for upscaling, ESPCN uses a novel approach called subpixel convolution, which not only enhances the resolution but also preserves image details and reduces computational complexity.

The ESPCN architecture comprises three key stages. Initially, feature extraction layers use convolutional filters (typically 3×3 in size) to extract relevant features from the low-resolution input image. These layers are critical for capturing the underlying patterns in the image data, essential for the subsequent upscaling process. Next, the network applies nonlinear mapping through activation functions such as the rectified linear unit (ReLU). This step introduces necessary nonlinearities into the system, enabling the network to learn complex mappings from low- to high-resolution data. The core of ESPCN is the subpixel convolution layer that rearranges the outputs of the previous layers from low-resolution space into a high-resolution output through a learned convolution followed by a periodic shuffling operation. This innovative approach allows for the upscaling of the image without traditional interpolation, resulting in sharper and more detailed images.

The transformation performed by ESPCN can be described by the following formula:

$$F^{\mathrm{SPM}} = P\left(S\left(W * F^{\mathrm{LR}} + b\right)\right) \tag{4}$$

where F^{LR} represents the low-resolution input image, and W and b denote the weights and biases of the convolutional layers, respectively. S is the shuffling operation that reorganizes the convolutional output into a high-resolution format. P includes any postprocessing steps such as clipping or additional filtering. This method significantly reduces the computational burden compared with traditional upscaling methods and provides a substantial improvement in the quality of the upscaled images.

It is important to emphasize that the subpixel mapping approach in this study differs from conventional endmember utilization methods. Rather than directly processing the endmembers themselves, this method primarily focuses on magnifying the endmember abundance maps through SR techniques. This novel approach allows for the enhanced representation of spatial details in hyperspectral images, facilitating more accurate and detailed analysis.

Finally, wavelet transformation is applied to effectively integrate spectral information obtained from the HASM method with spatial information introduced through unmixing in the subpixel mapping process. The wavelet transformation is particularly effective in processing hyperspectral data, as it allows for the decomposition of images at various scales, capturing detailed information at different resolutions. This scalability is crucial in isolating and enhancing specific features in both the spectral and spatial domains, which are often at different scales and require different treatments for optimal enhancement.

The process begins with the application of the HASM method to accurately map the spectral properties at a subpixel level, ensuring high resolution and precision in the representation of spectral data. Concurrently, ESPCN is used to enhance the spatial resolution of the hyperspectral image, focusing on increasing the clarity and detail of spatial features without compromising the spectral integrity.

The wavelet transform alters or modulates the scale size, thereby causing the family of fundamental functions within the wavelet transform to possess varying frequency band coverages for distinct information components.

For the fusion of images using wavelet decomposition, the discrete wavelet decomposition method is commonly used. The discrete wavelet decomposition is represented by the following equations:

$$\begin{cases} v_L\left[m, n'\right] = \displaystyle\sum_{q=0}^{Q-1} F\left[m, 2n' - q\right] l\left[q\right] \\ v_H\left[m, n'\right] = \displaystyle\sum_{q=0}^{Q-1} F\left[m, 2n' - q\right] h\left[q\right] \end{cases} \tag{5}$$

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Algorithm 1 WavT(F^{HASM}, F^{SPM})

Input: F^{HASM}, F^{SPM}

Output: F^{wav}

1: Wavelet decomposition.

a. Wavelet decomposition of images F^{HASM} and F^{SPM} (level 1).
$[\bar{F}^{HASM1}, \acute{F}^{HASM1}, \check{F}^{HASM1}, \grave{F}^{HASM1}] = wavedec(F^{HASM})$
$[\bar{F}^{SPM1}, \acute{F}^{SPM1}, \check{F}^{SPM1}, \grave{F}^{SPM1}] = wavedec(F^{SPM})$

b. Wavelet decomposition of the low-frequency images $\bar{F}^{HASM1}$ and $\bar{F}^{SPM1}$ (level 2).
$[\bar{F}^{HASM2}, \acute{F}^{HASM2}, \check{F}^{HASM2}, \grave{F}^{HASM2}] = wavedec(\bar{F}^{HASM1})$
$[\bar{F}^{SPM2}, \acute{F}^{SPM2}, \check{F}^{SPM2}, \grave{F}^{SPM2}] = wavedec(\bar{F}^{SPM1})$

2: Fusion rules.

a. According to the decomposed source images, fusion rules are designed for high-resolution images.
$\bar{F}^{wav2} \leftarrow B * \bar{Y}^{SPM2} + (1-B) * \bar{F}^{HASM2}$
$\acute{F}^{wav2} \leftarrow \acute{F}^{HASM2}$
$\check{F}^{wav2} \leftarrow \check{F}^{HASM2}$
$\grave{F}^{wav2} \leftarrow \grave{F}^{HASM2}$
$\acute{F}^{wav1} \leftarrow \acute{F}^{HASM1}$
$\check{F}^{wav1} \leftarrow \check{F}^{HASM1}$
$\grave{F}^{wav1} \leftarrow \grave{F}^{HASM1}$

3: Wavelet reconstruction.

a. Reconstruct the image based on the decomposed image obtained by the fusion rule (level 1).
$\bar{F}^{wav1} = waverec(\bar{F}^{wav2}, \acute{F}^{wav2}, \check{F}^{wav2}, \grave{F}^{wav2})$

b. Reconstruct the image based on the decomposed image obtained by the fusion rule (level 2).
$F^{wav} = waverec(\bar{F}^{wav1}, \acute{F}^{wav1}, \check{F}^{wav1}, \grave{F}^{wav1})$

$$\begin{cases} \bar{F}[m',n'] = \sum_{q=0}^{Q-1} v_L[2m'-q,n']l[q] \\ \acute{F}[m',n'] = \sum_{q=0}^{Q-1} v_L[2m'-q,n']h[q] \\ \check{F}[m',n'] = \sum_{q=0}^{Q-1} v_H[2m'-q,n']l[q] \\ \grave{F}[m',n'] = \sum_{q=0}^{Q-1} v_H[2m'-q,n']h[q] \end{cases} \tag{6}$$

where $l[q]$ and $h[q]$ represent the low-pass and high-pass filters, respectively. $F[m,n]$ denotes the original image, with m and n denoting the coordinate points corresponding to the matrix of the image conversion. $\mathrm{nu}_L[m,n]$ and $\mathrm{nu}_H[m,n]$ signify the results of filtering by $l[q]$ and $h[q]$, respectively. The wavelet transform primarily uses these filters to downconvert the image and extract its information

$$l(q) = (-1)^q h(2Q-1-q). \tag{7}$$

The relationship between $l[q]$ and $h[q]$ is shown in (7). Equations (5) and (6) represent 1-D and 2-D discrete wavelet decomposition, respectively.

Due to the multiscale analysis in wavelet transform theory, the low-pass filter in wavelet decomposition uses a large scale to isolate the low-frequency component of the image. This large-scale decomposition captures the overall information of the image. Conversely, the high-pass filter separates the high-frequency component of the image, revealing its finer details. As depicted in (6), $\bar{F}[m,n]$ represents the overall structure of the image, while $\acute{F}[m,n]$, $\check{F}[m,n]$, and $\grave{F}[m,n]$ capture its detailed features. The transformation equations (5) and (6) can be simplified as follows:

$$\left[\bar{F}, \acute{F}, \check{F}, \grave{F}\right] = \mathrm{wavedec}(F) \tag{8}$$

which indicates that when we input an image F, it can be decomposed into four different components $\bar{F}, \acute{F}, \check{F}, \grave{F}$ through a single wavelet transform. In addition, a new image can be reconstructed by applying the inverse operations of (5) and (6) to four different components

$$F = \mathrm{waverec}\left(\bar{F}, \acute{F}, \check{F}, \grave{F}\right). \tag{9}$$

A key feature of the wavelet transform for images is its ability to replace components or design fusion rules to integrate information from different inputs. It is important to note that in this work, the wavelet transform involves two levels of decomposition. Furthermore, considering the differences in information content among the images, we introduce a parameter B to constrain the information during fusion. B is an empirical value, and through testing, we found that setting B to 1/3 yields the best results.

By combining these two methodologies through wavelet transformation, it becomes feasible to generate an SR image that capitalizes on the advantages of each technique. Through wavelet transformation, the image is broken down into multiple frequency bands, allowing for individual processing tailored to each band's unique characteristics. The low-frequency bands, containing the bulk of the image's energy and representing large-scale structures, can be enhanced by incorporating the spatial details offered by ESPCN. Meanwhile, the high-frequency bands, responsible for capturing edges and intricate details, can benefit from the precise spectral mapping provided by HASM.

This dual-enhancement strategy guarantees that the SR image not only showcases improved spatial resolution but also showcases refined spectral features. This approach not only elevates the overall quality of the resultant image but also retains crucial attributes in both the spectral and spatial domains, achieving a feature-complementary SR that offers more comprehensive and precise information compared with methods used in isolation.

Using wavelet transformation, we can extract detailed spectral information about material composition and properties through the HASM method. Concurrently, the subpixel mapping process provides more accurate spatial distribution information. This integration of spectral and spatial data not only enhances the resolution of the final hyperspectral image

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but also improves the identification and comprehension of various features within complex urban environments. This comprehensive approach enables a more nuanced and detailed analysis, facilitating better decision-making and planning in urban management and development.

IV. Experiments and Analysis

A. Dataset Introduction

1) Pavia Hyperspectral Dataset: The Pavia hyperspectral dataset features urban imagery from the city of Pavia, Italy. These data were acquired by the PRISMA satellite and include 199 spectral bands covering the visible to near-infrared spectrum. Each image has a spatial resolution of 1024 × 1024 pixels, with each pixel size being approximately 1.3 m on the ground, providing detailed urban structure and material composition insights. The selected hyperspectral dataset comprises 199 spectral bands, providing a comprehensive coverage across the visible to near-infrared spectrum. Each spectral band contains 120 × 120 pixels, resulting in a granular high-resolution image that captures intricate details of the urban landscape. This high level of detail is crucial for applications such as land cover classification, vegetation detection, and urban infrastructure analysis.

Furthermore, the spatial resolution of each pixel allows for detailed analysis of urban features, which are essential for accurate mapping and monitoring of urban areas. The use of such high-resolution hyperspectral imagery enables the identification of materials and objects within the urban environment based on their spectral signatures, which differ significantly across the various bands of the dataset.

2) Washington DC Hyperspectral Dataset: The Washington DC hyperspectral dataset encompasses high-resolution imaging of the urban areas of Washington, DC, USA. The dataset includes 191 spectral bands, with a spectral coverage from 400 to 2500 nm. The images feature a spatial resolution of 1.5 × 1.5 m per pixel, which allows for detailed mapping of urban materials, land use, and environmental monitoring. This dataset is crucial for governmental and environmental applications focusing on urban management and development planning.

3) Berlin-Urban-Gradient Dataset (EnMAP): The Berlin-Urban-Gradient dataset, collected as part of EnMAP, offers hyperspectral imagery of Berlin, Germany. It comprises more than 240 spectral bands with a spectral range from 420 to 2450 nm. The dataset provides images at a spatial resolution of 30 × 30 m per pixel, suitable for analyzing urban landscapes, vegetation health, and pollution levels in both urban and peri-urban areas.

B. Evaluation Metrics

To assess the performance of various SR methods, we used several key evaluation metrics.

1) *Entropy:* Higher values indicate better performance. A higher entropy suggests that the image contains more information, implying better preservation of complexity and details in the super-resolved image.
2) *Spatial and Spectral Measures of Distortion:* Lower values indicate better performance. Spatial and spectral measures of distortion (SMD) measures the distortion in both the spatial and spectral domains, so a lower value means less distortion and better preservation of both the spatial and spectral characteristics.
3) *Edge-Aware Variance:* Higher values indicate better performance. Edge-aware variance (EAV) focuses on edge sharpness and clarity, which are crucial in urban settings. A higher EAV suggests better preservation of fine details and edge information.
4) *Correlation Coefficient:* Higher values indicate better performance. Correlation coefficient (CC) ranges from −1 to 1, with 1 indicating perfect positive correlation. In our context, a higher CC means the super-resolved image more accurately reproduces the original image.
5) *Spectral Angle Mapper:* Lower values indicate better performance. Spectral angle mapper (SAM) measures the angular difference between spectral vectors of the original and super-resolved images. A lower SAM value suggests higher spectral similarity and better preservation of spectral integrity.

C. Experimental Comparison

For the comparative analysis, we evaluated several classic SR methods that do not require training datasets. These methods included bicubic interpolation, edge-directed [31], edge-guided [32], autoregressive [37], and multiscale [38]. Our choice was driven by the need to assess SR performance in the absence of ground-truth data, which is a common challenge in remote sensing applications.

1) Pavia Hyperspectral Dataset: The experimental results are meticulously presented in Fig. 2 and Table I. Upon closer examination of the magnified areas within Fig. 2, it becomes evident that certain methods such as bicubic interpolation and autoregressive yielded suboptimal results, characterized by visible block artifacts. These artifacts not only degrade the visual esthetics of the images but also signify a significant loss of fine details and edge definition, which are crucial for accurate interpretation and analysis of urban structures.

Conversely, methods such as edge-directed and multiscale demonstrated superior performance with regard to preserving the integrity of edges and details. The edge-directed method, in particular, was effective in enhancing edge definition using advanced edge-detection algorithms that adaptively refine the image's edges, thus preventing the blurring commonly seen with simpler methods. The Multiscale approach, leveraging varying scales of image features, successfully preserved detailed textures and contours that are typically lost in lower quality SR processes.

Numerical evaluations further underscore the superior performance of our proposed method. Specifically, our method achieved the highest scores in terms of information entropy among all the methods tested. High entropy values indicate a rich presence of spatial details within the image, suggesting that our method was particularly effective in introducing and preserving complex image structures and variations within the

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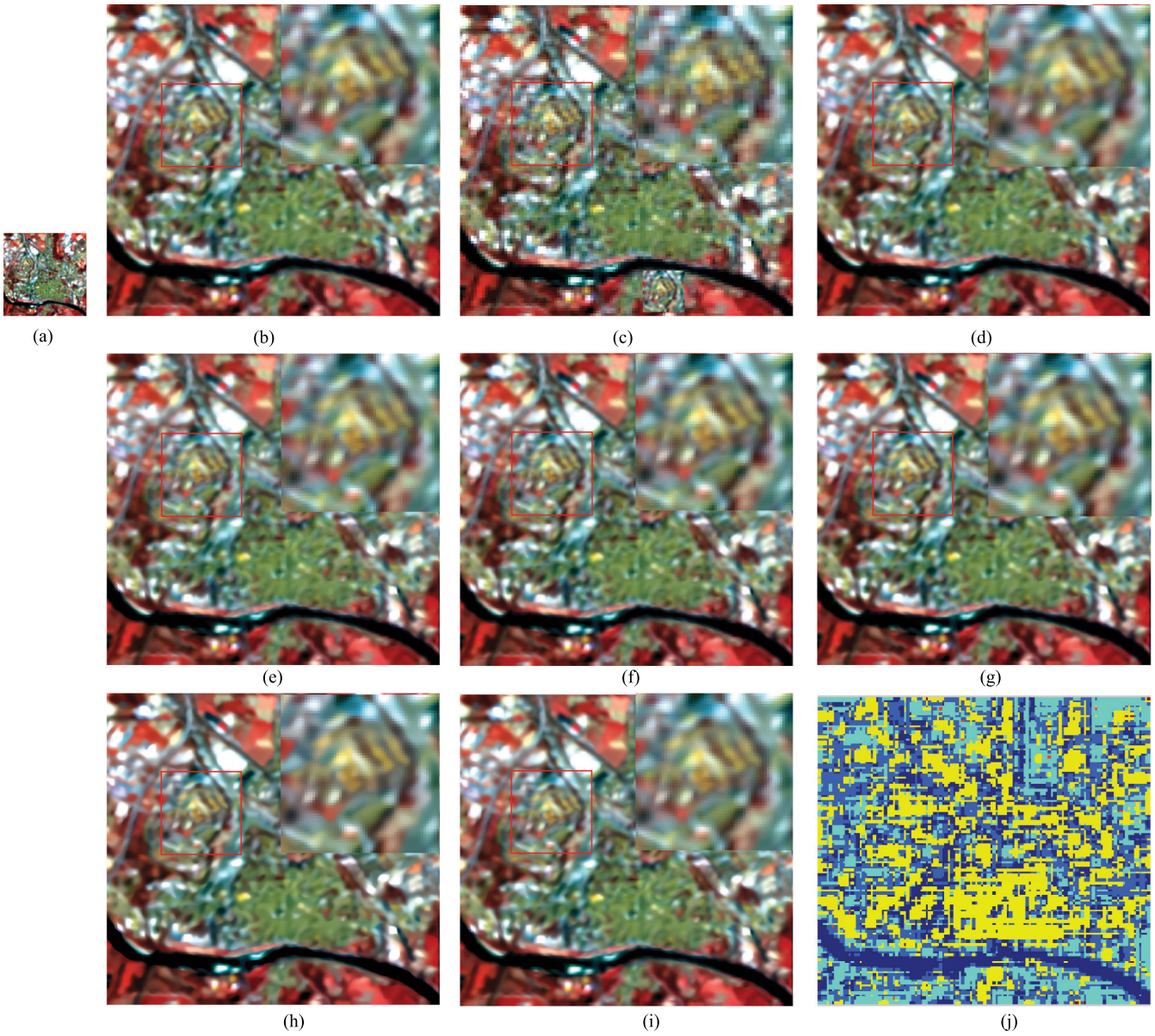


Fig. 2. Experimental result. (a) Low-resolution image. (b) HASM. (c) Bicubic. (d) Edge-directed. (e) Edge-guided. (f) Autoregressive. (g) Multiscale. (h) HASM + subpixel mapping. (i) Our method. (j) Subpixel mapping.

TABLE I
EXPERIMENTAL RESULT OF THE PAVIA DATASET

Result	Bicubic	Edge-directed	Edge-guided	auto regressive	Multiscale	Ours
Entropy	22.1462	22.1749	25.3472	27.2146	29.3670	**30.1484**
SMD	14.2351	15.2357	17.4234	17.3412	17.7319	**18.2557**
EAV	74.1273	76.2371	76.2478	78.2361	82.1369	**82.1462**

super-resolved images. This is a crucial attribute for applications that depend on the nuanced interpretation of spatial features.

Moreover, SMD results indicate that our method not only excelled in enhancing spatial details but also maintained high spectral fidelity. This dual capability ensures that the spectral characteristics of the urban landscape, such as material properties and subtle color variations, are accurately represented, which is essential for tasks such as material identification and land cover classification.

In addition, EAV, our method, demonstrated exemplary performance in edge representation. EAV is a critical measure as it assesses the ability of the SR process to enhance and preserve the clarity of edges—a vital factor in delineating distinct urban elements such as buildings, roads, and vegetation boundaries. Superior edge representation facilitates better segmentation and sharper delineation, enabling more precise urban planning and monitoring.

In summary, the comprehensive performance of our method across multiple metrics—entropy, SMD, and EAV—highlights

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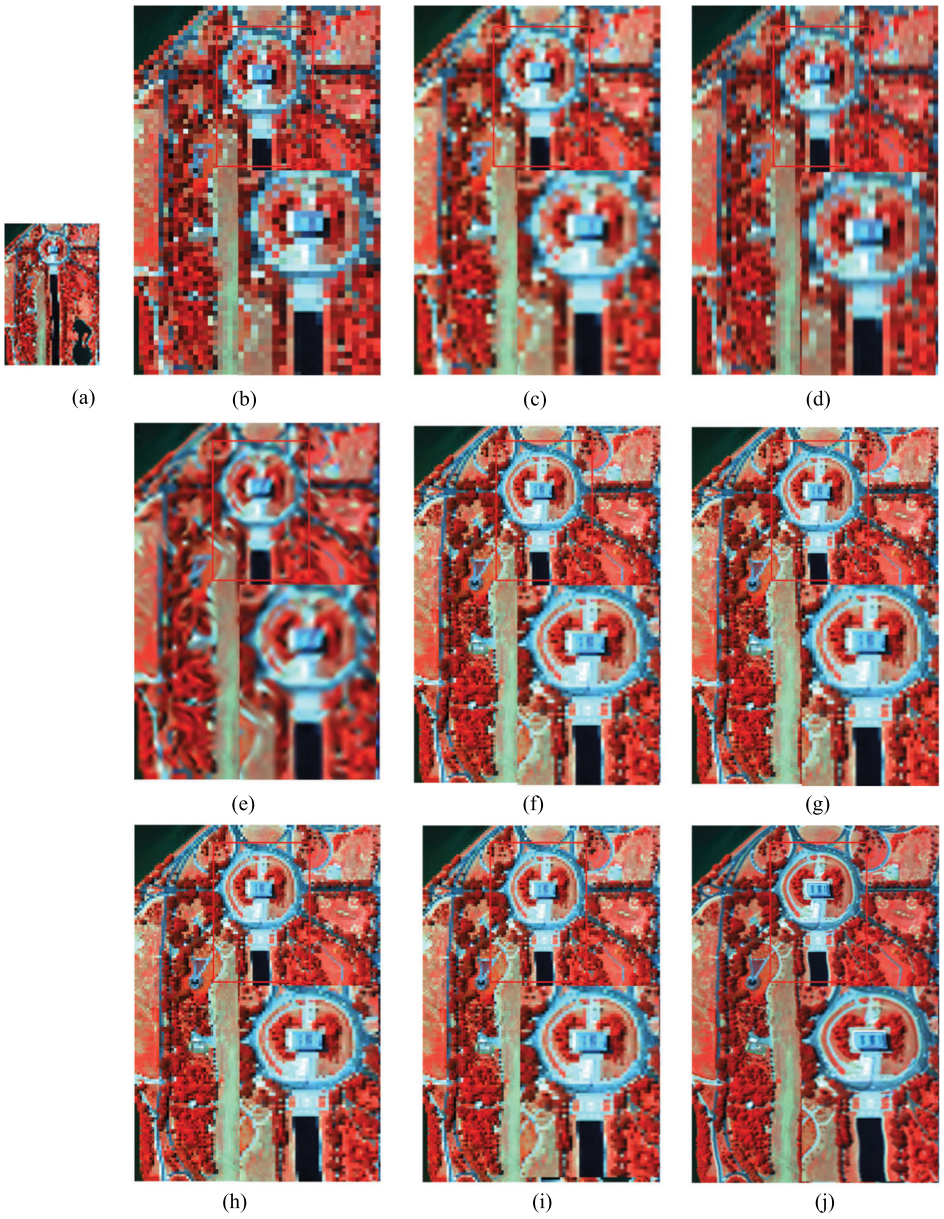


Fig. 3. Experimental result. (a) Low-resolution image. (b) HASM. (c) Bicubic. (d) Edge-directed. (e) Edge-guided. (f) Autoregressive. (g) Multiscale. (h) HASM + subpixel mapping. (i) Our method. (j) Ground truth.

its efficacy in enhancing both the spatial and spectral qualities of urban hyperspectral images. The ability to effectively balance the enhancement of fine details with the preservation of spectral integrity positions our method as particularly advantageous for advanced urban analysis and remote sensing applications.

This comprehensive analysis highlights the efficacy of our approach in enhancing both the spatial and spectral qualities of urban hyperspectral images, making it particularly useful for applications in urban planning and environmental monitoring.

2) Washington DC Hyperspectral Dataset: The analysis of experimental results in Fig. 3 and Table II reveals distinct performance advantages of our method. Notably, it achieves the highest entropy value, indicating an exceptional ability to capture detailed image information. Similarly, it leads in preserving spatial and spectral details, as evidenced by the highest Spatial and SMD. Furthermore, our method slightly outperforms the advanced multiscale method in terms of EAV, showcasing superior edge preservation. In addition, it records the highest CC, demonstrating that the super-resolved images maintain close alignment with the original in terms of overall structure. Finally, our method significantly reduces the SAM values, ensuring minimal spectral distortion and enhanced spectral fidelity.

These results clearly demonstrate that our SR method surpasses traditional techniques such as bicubic, edge-directed, edge-guided, autoregressive, and even advanced multiscale methods across all the evaluated metrics. The superior performance in maintaining high detail, integrity, and fidelity of urban hyperspectral images positions our method as particularly suitable for critical applications in urban planning, environmental monitoring, and remote sensing. This robust

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TABLE II
PERFORMANCE METRICS OF DIFFERENT SR METHODS ON URBAN HYPERSPECTRAL DATA

Result	Bicubic	Edge-directed	Edge-guided	Autoregressive	Multiscale	Ours
Entropy	22.1313	23.7453	24.1358	26.2175	28.5784	32.1592
SMD	14.1215	14.2123	14.5242	15.1153	18.1264	20.4234
EAV	74.1574	76.72351	82.3588	84.1238	84.7291	85.1346
CC	0.8127	0.8254	0.8568	0.8721	0.9125	0.9523
SAM	0.147	0.0952	0.0921	0.0831	0.0623	0.0526

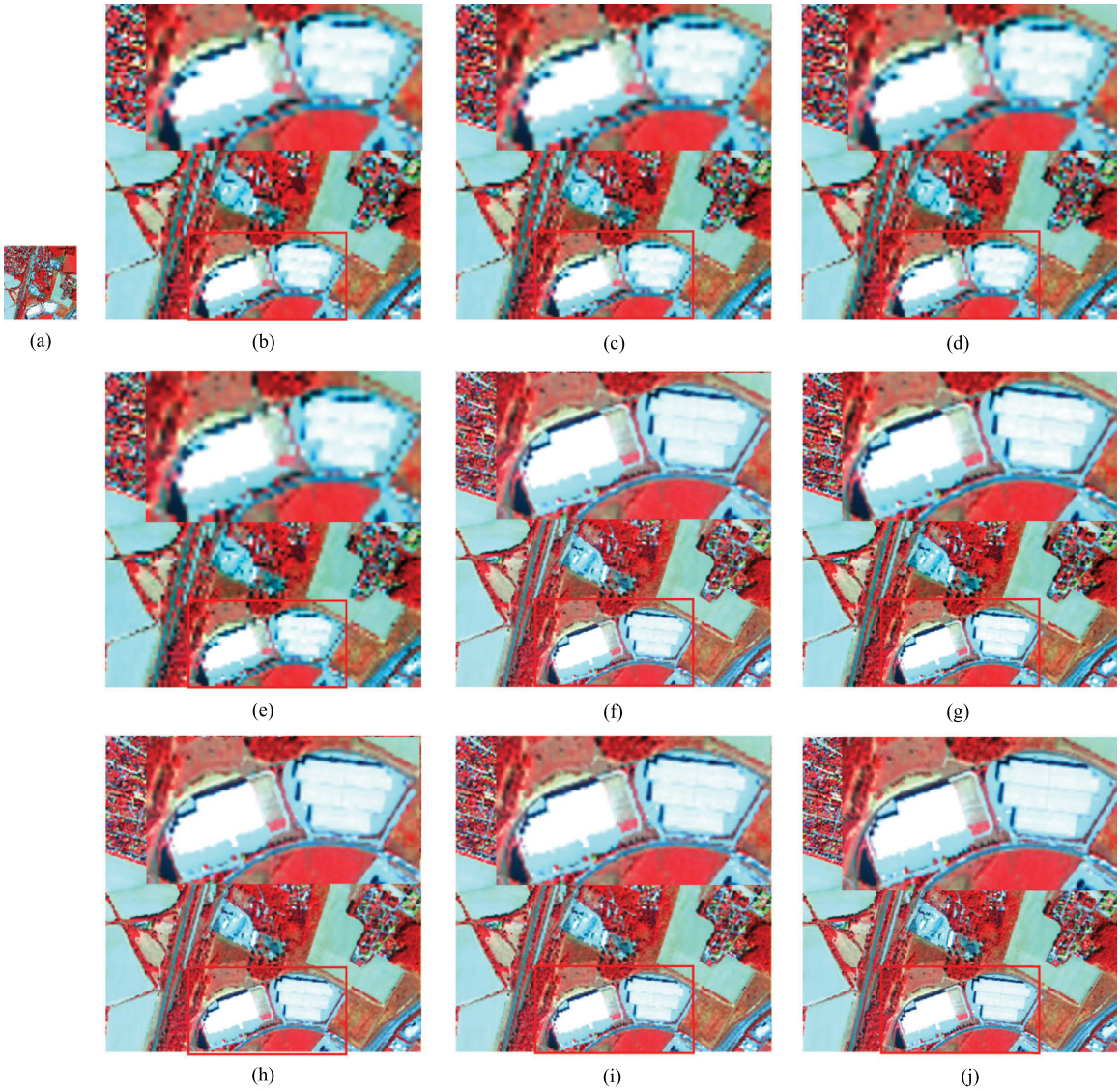


Fig. 4. Experimental result. (a) Low-resolution image. (b) HASM. (c) Bicubic. (d) Edge-directed. (e) Edge-guided. (f) Autoregressive. (g) Multiscale. (h) HASM + subpixel mapping. (i) Our method. (j) Ground truth.

TABLE III
PERFORMANCE METRICS OF DIFFERENT SR METHODS ON URBAN HYPERSPECTRAL DATA

Result	Bicubic	Edge-directed	Edge-guided	Autoregressive	Multiscale	Ours
Entropy	24.1373	25.2751	26.1238	31.2187	32.1274	34.2590
SMD	15.1263	15.2742	15.3942	18.1235	20.8216	22.1524
EAV	77.1236	77.7512	80.1478	84.1268	85.1695	86.1263
CC	0.8237	0.8438	0.8479	0.9141	0.9317	0.9523
SAM	0.0791	0.0747	0.0621	0.0461	0.0427	0.0316

performance underscores the significant potential of our approach to enhance the utility of hyperspectral imaging in urban analysis, providing a powerful tool for detailed environmental and urban assessments.

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3) Berlin-Urban-Gradient Dataset (EnMAP): The analysis of the results in Fig. 4 and Table III reveals that our method exhibits superior performance across various metrics compared with other SR methods on urban hyperspectral data. First, in terms of entropy, our method achieved the highest value (34.2590), indicating the highest information content present in the super-resolved images. This suggests that our method effectively captures and preserves the intricate details within the data.

Second, concerning structural similarity, our method again outperforms others, as indicated by the Structural SMD. With a value of 22.1524, our method demonstrates better preservation of structural features compared with alternative techniques. This implies that our method produces super-resolved images that closely resemble the original data.

Moreover, our method excels in maintaining the statistical properties of the data, as evidenced by the highest EAV of 86.1263. This indicates that our method successfully preserves the statistical characteristics of the urban hyperspectral data during the SR process.

Furthermore, our method achieves the highest CC of 0.9523, implying a strong linear relationship with the ground-truth data. This highlights the accuracy and fidelity of our method in reproducing the original spectral information.

Finally, our method demonstrates the lowest SAM value of 0.0316, indicating the smallest spectral angle difference from the ground truth. This suggests that our method accurately reconstructs the spectral signatures of the urban hyperspectral data.

V. Conclusion

In this article, with the objective of obtaining fast, spectrally faithful, and spatially detailed urban hyperspectral images, we have proposed a novel urban hyperspectral image SR method that capitalizes on the advantages of both the interpolation and subpixel mapping techniques. The experimental results demonstrate that our proposed method offers significant improvements over classical approaches. It successfully meets the anticipated goals in terms of limited computational requirements, spectral fidelity of the results to the original low-resolution dataset, while, as expected, enhancing the spatial details.

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Jia Chen (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from China University of Geosciences, Wuhan, China, in 2015, 2017, and 2022, respectively.

He is currently a Post-Doctoral Researcher with the University of Pavia, Pavia, Italy. His research interests include hyperspectral image unmixing, image fusion, machine learning, and deep learning.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisbon, Lisbon, Portugal, in 2011.

She is a Full Professor with China University of Geosciences, Wuhan, China. Her main research interests comprise remotely sensed hyperspectral image analysis, signal processing, supervised/semi-supervised learning, and active learning.

Dr. Li is the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been a Guest Editor of several journals, including PROCEEDINGS OF THE IEEE and *ISPRS Journal of Photogrammetry and Remote Sensing*.



Paolo Gamba (Fellow, IEEE) received the Laurea (cum laude) and Ph.D. degrees in electronic engineering from the University of Pavia, Pavia, Italy, in 1989 and 1993, respectively.

He is currently a Professor with the Telecommunications and Remote Sensing Laboratory, University of Pavia. He has been invited to give keynote lectures and tutorials on several occasions about urban remote sensing, data fusion, EO data for physical exposure, and risk management. He has published more than 180 articles in international peer-reviewed journals and presented 320 research works in workshops and conferences.

Dr. Gamba was elected in the GRSS AdCom in 2014, served as the GRSS President from 2019 to 2020, and is currently the GRSS Past President. He has served as the Chair for the Data Fusion Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) from October 2005 to May 2009. He was the Organizer and the Technical Chair of the biennial GRSS/ISPRS Joint Workshops on "Remote Sensing and Data Fusion Over Urban Areas" from 2001 to 2015. He has served as the Technical Co-Chair of the 2010, 2015, and 2020 IGARSS conferences, in Honolulu, HI, USA; Milan, Italy; and on-line, respectively. He has also served as the Editor-in-Chief for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2009 to 2013.

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